MIDTERM EXAMINATION

STA301- Statistics and Probability (Session - 1)

Question No: 1 (Marks: 1) - Please choose one

If any value in the data is zero, then it is not possible to have:

- ► Harmonic Mean
- ► Arithmetic Mean
- ► Median

Mode

- ► Mode

Question No: 2 (Marks: 1) - Please choose one

For a symmetrical distribution………..is equidistance from median:

- ► X_0 and X_m
- ► Q_1 and Q_3
- ► X_0 and Q_1
- ► X_m and Q_3

Question No: 3 (Marks: 1) - Please choose one

Which one of the following measure is not used in ‘five number summery’:

- ► X_0
- ► Q_3

► $\bar{X}$

► Q_1

Question No: 4 (Marks: 1) - Please choose one

If Mean = 25 & S.D is 5 then C.V is

► 100%

► 25%

► 20%

► 10%

Question No: 5 (Marks: 1) - Please choose one

A coin is tossed 3 times then, the number of sample points ii the sample space is:

► 3

► 8

► 6

► 4

Question No: 6 (Marks: 1) - Please choose one

What is the difference between a permutation and a combination:

► In a permutation order is important and in a combination it is not

► In a permutation order is not important and in a combination it is important

► A combination is based on the classical definition of probability

►

Question No: 7 (Marks: 1) - Please choose one

What we consider in simple correlation analysis:

► Several independent variables

► Strength of the association between two variables

► Intercept with the X-axis

►

Question No: 8 (Marks: 1) - Please choose one

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient ‘r’:

► $0 = r = 1$

► $r = 0$

► $r = +1$

► $r=-1$

Question No: 9 (Marks: 1) - Please choose one

Which of the following measure of dispersion is least affected by extreme values of observations in a data:

► Range

► Quartile deviation

► Mean absolute deviation

► Standard deviation

Question No: 10 (Marks: 1) - Please choose one

The approximate number of observations in a set of data covered by the interval, $Median \pm Q.D$ are:

► 50 per cent

► 68.5 per cent

► 95.4 per cent

► 99 per cent

Question No: 11 (Marks: 1) - Please choose one

The dispersion which is calculated from all observed values is:

► Standard deviation

► Quartile deviation

► Rang

► Coefficient of Rang

Question No: 12 (Marks: 1) - Please choose one

In a set of 20 values all the values are 2, what is the value of Geometric Mean:

► 2

► 5

► 10

► 20

Question No: 13 (Marks: 1) - Please choose one

In a set of 10 values all the values are 5, what is the value of 5^{th} Decile:

► 2

► 5

► 10

► 20

Question No: 14 (Marks: 1) - Please choose one

If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

► W.M

► H.M

► A.M

► G.M

Question No: 15 (Marks: 1) - Please choose one

In stem and leaf plot, data is measured on:

- ▶ Ratio Scale
- ▶ Interval Scale
- ▶ Ordinal scale
- ▶ Nominal Scale

Question No: 16 (Marks: 1) - Please choose one

In statistics,we deal with:

- ▶ Individuals
- ▶ Isolated items
- ▶ Aggregates of facts
- ▶ Qualitative data

Question No: 17 (Marks: 1)

What is a sample?

Question No: 18 (Marks: 1)

From the given table, write down the stem values.

(Leading Digit)	(Trailing Digit)
1	8 2
2	9 6 7
3	1 7 9 0 5
4	8 3 0 2 8 9
5	4 1 2 3 7 8
6	4 1 5 2 7 8
7	1 4

Question No: 19 (Marks: 2)

Distinguish among *nominal*, *ordinal* levels of measurement:

Question No: 20 (Marks: 3)

Elaborate cumulative frequency.

Question No: 21 (Marks: 5)

The probability that a student passes mathematics is 2/3 and the probability that he passes English is 4/9. If the probability of passing at least one course is 4/5, what is the probability that he will pass both courses?

Question No: 22 (Marks: 10)

A survey of 50 retail establishments had assistants, excluding proprietors, as follows:
2,3,9,0,4,4,1,5,4,8,5,3,6,6,
0,2,7,6,4,8,4,3,3,1,0,8,7,5,
1,3,4,7,2,4,7,5,2,6,3,1,7,5,
4,6,4,2,5,3,4.
Arrange the values as frequency distribution.